

Jonathan Ma

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EDUCATION

Johns Hopkins University

Master of Science in Engineering, Applied Mathematics and Statistics

Thesis: *Calibration-Aware Recurrent Neural Networks for Multivariate Probabilistic Forecasting under Distribution Shift*

Baltimore, MD

Expected Dec 2026

Rutgers University

Bachelor of Science in Finance; Bachelor of Arts in Economics; Minor in Statistics

Honors: Summa Cum Laude, Honors in Economics

Camden, NJ

May 2025

RESEARCH & ANALYTICS EXPERIENCE

University of Utah, Sharon Eccles Steele Center for Translational Medicine

Affiliate Researcher

Salt Lake City, UT

Jun 2026 – Present

- Built computational pipelines to quantify retinal OCT biomarkers by integrating HEYEX segmentations, engineered imaging features, and logistic regression-based weighting for progression-risk modeling.
- Evaluated classical image analysis and ResNet50 transfer-learning approaches on 24,000+ OCT images, achieving approximately 93% held-out classification accuracy and using Grad-CAM to assess model behavior.
- Performed bootstrap validation, ROI perturbation, and segmentation sensitivity analyses while collaborating with clinicians and biostatisticians on interpretable, HIPAA-compliant workflows.

Rutgers University, Economics Research

Research Assistant

Camden, NJ

Jan 2023 – Dec 2025

- Built a Python and R pipeline to collect, preprocess, and quality-check 7,500+ earnings-call and conference transcripts for financial NLP and disclosure analysis.
- Developed a FinBERT-assisted sentiment workflow with structured prompting and manual quality assurance to improve consistency across thousands of financial documents.
- Constructed a database of 23,500+ equity analysts and developed a Selenium-assisted entity-resolution pipeline that reduced 3,750 ambiguous records to 54 unresolved cases.
- Developed generalized linear mixed models and structural equation models using Youth Risk Behavior Survey data to evaluate behavioral, demographic, and socioeconomic determinants of adolescent health outcomes.
- Benchmarked economics curricula at peer universities and synthesized a qualitative review with current students into recommendations for curriculum development, outreach, and program enrollment.

SELECTED PROJECTS & PUBLICATIONS

CA-RNN: Calibration-Aware Probabilistic Forecasting (GitHub)

- Developed and evaluated recurrent and Bayesian recurrent models with full-covariance predictive distributions and differentiable calibration penalties across simulated, market, and distribution-shift settings.
- Packaged the model as a reproducible PyTorch library with deterministic training, leakage-safe splits, structured experiment logs, validation checkpointing, automated tests, and documented evaluation artifacts.

Modern AutoCriteria: Clinical-Trial Eligibility Extraction (GitHub)

- Built modular LLM extraction and validation workflows with clinical terminology normalization, structured logging, token monitoring, automated retries, and model-integration tests, achieving 91.8% Exact Match and 0.96 F1.
- Developed a clinically aware distance metric combining human judgment with LLM-as-a-Judge evaluation to distinguish lexically similar but contradictory eligibility criteria.

Peer-Reviewed Publication: Zhu, S., Ma, J., and Liu, Z. (2026). “Bayesian Hierarchical Modeling and Clustering for Malignant Cancer Diagnosis.” *Argumentation Based Systems Journal*. (DOI)

TECHNICAL SKILLS

Python: PyTorch, scikit-learn, scikit-fda, CVXPY, PyMC, pandas, NumPy, sentence-transformers, OpenCV, Selenium, bs4

R: tidyverse, dplyr, ggplot2, CARET, lavaan, brms, bvar, TSA, BART, sqldf

Additional Languages: SQL, Julia, JavaScript, MATLAB

CERTIFICATIONS

Bloomberg Market Concepts

2022

CFI Financial Modeling & Valuation Analyst (FMVA)

2024

CITI Good Clinical Practice

2026